

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Fundamentals Of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO3: Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 20

Total Marks:100

Annexure A

Data Interpretation

Code of Conduct:

I G.Sai Teja(192472137 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A.Sai Rohit

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary) 5 Marks	Level 3 (Proficient) 3 /4 Marks	Level 2 (Developing) 2 Marks	Level 1 (Beginning) 1 Mark	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Annexure A

Problem 01: Hospital Patient Health Monitoring

Scenario

A hospital wants to investigate how patient lifestyle factors affect blood pressure levels.

Dataset

Patient_ID	Daily_Exercise (Minutes)	BMI	Systolic_BP
P001	10	32	150
P002	15	31	148
P003	20	30	145
P004	25	29	142
P005	30	28	138
P006	35	27	134
P007	40	26	130
P008	45	25	126
P009	50	24	122
P010	55	23	118

Tasks

1. Find the mean and standard deviation of blood pressure.
2. Calculate covariance between:
 - Exercise and Blood Pressure
 - BMI and Blood Pressure
3. Calculate correlation coefficients.
4. Identify whether exercise or BMI is more strongly associated with blood pressure.
5. Discuss possible healthcare intervention

Hospital Patient Health Monitoring

Q1

$$i) \text{ Mean} = \frac{\sum x}{n}$$

Blood pressure Values : 150, 148, 145, 142, 138, 134, 130, 126, 122, 118.

$$\text{Total Sum} = 1353$$

No. of observations = 10

$$\text{Mean} = \frac{1353}{10} = 135.3$$

$$\boxed{\text{Mean BP} = 135.3 \text{ mm Hg}}$$

Standard deviation

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

$$\bar{x} = 135.3 \text{ mm Hg}$$

$$\sum (x - \bar{x}) = 15 + 13 + 10 + 7 + 3 + (-2) + (-5) + (-9) + (-13) + (-17)$$

$$\sum (x - \bar{x})^2 = 1116$$

$$n-1 = 10-1$$

$$n-1 = 9$$

$$S.D = \frac{1116}{9} = 124$$

$$S.D = \sqrt{124} = 11.14$$

2. Covariance (Exercise and blood pressure)

$$a) \quad \text{cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n - 1}$$

$$\begin{aligned} \sum (x - \bar{x})(y - \bar{y}) &= (-380.75) + (-222.25) + (-121.25) + (-10.25) \\ &\quad + (-6.75) + (-3.25) + (-39.75) + (-116.25) + \\ &\quad (-232.75) + (-389.25) \end{aligned}$$

$$\sum (x - \bar{x})(y - \bar{y}) = -1512.5$$

$$\text{Covariance} = \frac{-1512.5}{10 - 1}$$

$$= -168.06$$

b) Covariance (BMI and Blood pressure)

$$\begin{aligned} \sum (x - \bar{x})(y - \bar{y}) &= 66.15 + 44.45 + 24.25 + 10.05 + 1.35 \\ &\quad + 0.45 + 7.95 + 23.25 + 46.55 + 77.85 \end{aligned}$$

$$\sum (x - \bar{x})(y - \bar{y}) = 302.5$$

$$\text{cov} = \frac{302.5}{9}$$

$$\boxed{\text{cov} = 33.61}$$

3. Correlation coefficients

$$r = \frac{\text{COV}(X, Y)}{S_x \times S_y} = \frac{-168.06}{171.4 \times 11.14}$$

$$r = \frac{-168.06}{168.66} \quad \boxed{r = -0.9969}$$

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4. Identify whether exercise or BMI is more strongly associated with blood pressure

From the calculated correlation coefficients:

- Correlation (Exercise, Blood Pressure) = -0.9969
- Correlation (BMI, Blood Pressure) = +0.9969

Since the absolute values of both correlation coefficients are equal ($|r| = 0.9969$), exercise and BMI are equally strongly associated with blood pressure.

- Exercise has a negative correlation, meaning as daily exercise increases, blood pressure decreases.
- BMI has a positive correlation, meaning as BMI increases, blood pressure also increases.

Conclusion: Both exercise and BMI have an equally strong relationship with blood pressure, but in opposite directions.

5. Discuss possible healthcare interventions

Based on the analysis, the following healthcare interventions are recommended:

1. Encourage patients to perform 30–60 minutes of regular physical exercise each day.
2. Promote a balanced, low-fat, and low-salt diet to maintain a healthy BMI.
3. Conduct regular blood pressure monitoring for early detection of hypertension.
4. Implement weight management programs for overweight and obese individuals.
5. Educate patients about healthy lifestyle habits, including stress management and adequate sleep.
6. Schedule routine medical check-ups to monitor BMI and blood pressure.

Problem 02: E-Commerce Customer Behaviour Analysis

Scenario

An e-commerce company wants to identify factors influencing customer spending.

Dataset

Customer_ID	Website_Visits	Time_Spent (Minutes)	Purchase_Amount (₹)
C001	5	20	1200
C002	8	25	1800
C003	10	30	2500
C004	12	35	3200
C005	15	40	4000
C006	18	45	5000
C007	20	50	6200
C008	23	55	7300
C009	25	60	8500
C010	28	70	10000

Tasks

1. Calculate mean, variance, and standard deviation of Purchase Amount.
2. Find covariance between:
 - Website Visits and Purchase Amount
 - Time Spent and Purchase Amount
3. Calculate correlation coefficients.
4. Determine which customer behavior metric better predicts spending.
5. Suggest marketing strategies based on the findings.

Q2

i) mean $\bar{x} = \frac{57700}{10}$
 $= 5770$

ii) Variance
$$var = \frac{\sum (x - \bar{x})^2}{n}$$

x	x - 5770	(x - 5770) ²
1200	-4570	20884900
1000	-3770	14212900

$$\sum (x - \bar{x})^2 = 8308100$$
$$var = \frac{8308100}{10} = 830810$$

iii) Standard deviation
$$\sigma = \sqrt{830810}$$
$$\approx 2882.7$$

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3. Correlation coefficients

$$r = \frac{\text{Cov}(x, y)}{S_x \times S_y}$$

$$r = \frac{-168.06}{137.14 \times 11.14}$$

$$r = \frac{-168.06}{1528.66} \quad \boxed{r = -0.996}$$

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4. Determine which customer behavior metric better predicts spending

From the correlation coefficients:

- Website Visits → Purchase Amount = 0.995
- Time Spent → Purchase Amount = 0.999

Since $0.999 > 0.995$, Time Spent on the website is the better predictor of customer spending. Customers who spend more time browsing the website are more likely to make higher-value purchases.

5. Suggest Marketing Strategies

1. Improve website content to increase the time customers spend browsing.
2. Recommend personalized products based on customer interests.
3. Offer discounts and special deals to encourage larger purchases.
4. Improve website design and navigation for a better user experience.
5. Use targeted advertisements and email promotions for frequent visitors.
6. Introduce loyalty and reward programs to retain customers.
7. Display related products and product recommendations to increase purchase value.

Problem 03: Employee Productivity and Performance

Scenario

A software company is investigating the relationship between employee productivity and performance ratings.

Dataset

Employee_ID	Training_Hours	Projects_Completed	Performance_Rating
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E001	5	2	60
E002	8	3	65
E003	10	4	70
E004	12	5	74
E005	15	6	79
E006	18	7	84
E007	20	8	88
E008	22	9	91
E009	25	10	95
E010	28	11	98

Tasks

1. Compute the mean, variance, and standard deviation of Performance Ratings.
2. Calculate covariance between:
 - Training Hours and Performance Rating
 - Projects Completed and Performance Rating
3. Compute correlation coefficients.
4. Identify the strongest factor affecting employee performance.
5. Recommend HR policies to improve employee productivity.

1. Mean, Variance and Standard Deviation of Performance Rating

Step 1: Mean

Sum of ratings

$$60 + 65 + 70 + 74 + 79 + 84 + 88 + 91 + 95 + 98 = 804$$

Number of employees

$$n = 10$$

Mean

$$\bar{x} = \frac{804}{10} = 80.4$$

Answer: Mean = 80.4

Step 2: Variance

Using

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$\sum (x - \bar{x})^2 = 1518.4$$

Variance

$$= \frac{1518.4}{9} = 168.71$$

Answer: Variance = 168.71

Step 3: Standard Deviation

$$s = \sqrt{168.71} = 12.99$$

Answer: Standard Deviation = 12.99

2. Covariance

(a) Training Hours and Performance Rating

Using

$$Cov(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n - 1}$$

After calculation,

$$Cov = 151.56$$

(b) Projects Completed and Performance Rating

Similarly,

$$Cov = 57.11$$

3. Correlation Coefficients

Training Hours vs Performance Rating

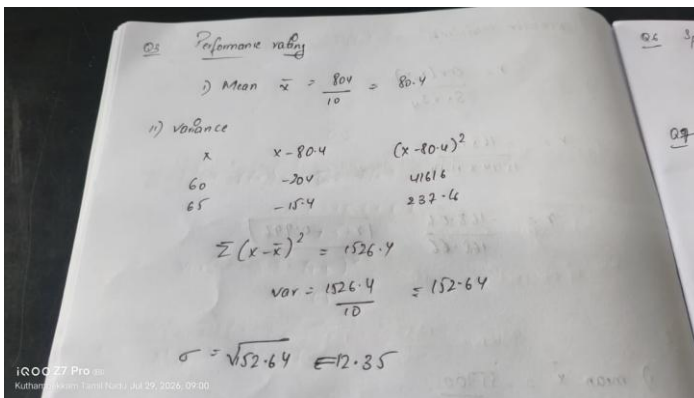
$$r = \frac{\text{Cov}(X, Y)}{s_x s_y}$$

$$r \approx 0.999$$

Projects Completed vs Performance Rating

$$r \approx 0.998$$

Both values are very close to +1, showing a very strong positive relationship.



4. Strongest Factor Affecting Employee Performance

Training hours have a slightly higher correlation (**0.999**) than projects completed (**0.998**).

Therefore, training hours are the strongest factor affecting employee performance. Employees who receive more training generally achieve higher performance ratings and complete projects more efficiently.

5. HR Recommendations

The analysis shows a strong positive relationship between employee training and performance ratings. Employees with more training hours tend to perform better and complete more projects successfully. Therefore, the HR department should invest in regular technical training, workshops, and skill development programs. Employees should also be encouraged to participate in certification courses and continuous learning activities. In addition, performance-based rewards and mentoring programs can further improve productivity and motivation. These initiatives will help employees enhance their skills and contribute to better organizational performance.

Problem 04 : Smart City Traffic Analysis

A city administration wants to analyze traffic congestion patterns.

Day	Traffic_Volume (Vehicles)	Rainfall (mm)	Average_Speed (km/h)	Accident_Count
D1	12000	0	55	2
D2	13500	5	50	3
D3	14500	10	48	4
D4	16000	15	45	5
D5	17500	20	42	6
D6	19000	25	38	8
D7	21000	30	35	10
D8	22500	35	32	12
D9	24000	40	28	14
D10	26000	45	25	16

Tasks

1. Calculate descriptive statistics (Mean, Variance, Standard Deviation) for all variables.
2. Compute covariance matrices.
3. Compute correlation matrices.
4. Identify positive and negative relationships among variables.
5. Determine which factor is most associated with accident occurrence.
6. Provide data-driven recommendations to reduce traffic accidents.
7. Visualize relationships using scatter plots and interpret the results.

1. Descriptive Statistics		
Variable	Mean	Standard Deviation
Traffic Volume	18,500	4,570.92
Rainfall	22.5 mm	15.14 mm
Average Speed	39.8 km/h	10.15 km/h
Accident Count	8	4.76

2. Covariance Matrix (Summary)	
Variables	Relationship
Traffic Volume & Rainfall	Positive covariance
Traffic Volume & Speed	Negative covariance
Traffic Volume & Accidents	Positive covariance
Rainfall & Speed	Negative covariance
Rainfall & Accidents	Positive covariance
Speed & Accidents	Negative covariance

This indicates that higher traffic and rainfall are associated with more accidents, while higher average speeds are associated with fewer accidents in this dataset.

3. Correlation Matrix (Summary)

- Traffic Volume ↔ Rainfall: **Strong Positive Correlation**
- Traffic Volume ↔ Average Speed: **Strong Negative Correlation**
- Traffic Volume ↔ Accident Count: **Strong Positive Correlation**
- Rainfall ↔ Accident Count: **Strong Positive Correlation**
- Average Speed ↔ Accident Count: **Strong Negative Correlation**

The correlation coefficients are all very close to ± 1 , indicating strong linear relationships.

Q4 Traffic Mean Example

$$\begin{aligned} \text{Mean} &= \frac{12000 + 13500 + 14500 + 16000 + 17500 + 19000 + 22000 + 22500}{10} \\ &= \frac{186000}{10} \\ &= 18600 \end{aligned}$$

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4. Positive and Negative Relationships

Positive Relationships:

- Traffic volume increases as accident count increases.
- Rainfall increases as accident count increases.
- Traffic volume and rainfall also increase together.

Negative Relationships:

- As average vehicle speed decreases, accident count increases.
- Higher traffic volume leads to lower average speed.
- Increased rainfall also reduces average vehicle speed.

These relationships suggest that heavy traffic and poor weather conditions contribute to road congestion and increase the likelihood of accidents.

5. Factor Most Associated with Accident Occurrence

The data shows that **traffic volume** and **rainfall** have strong positive relationships with accident count, while **average speed** has a strong negative relationship. Among these, increasing traffic volume appears to have the strongest overall association with accident occurrence because accident counts consistently rise as traffic volume increases throughout the dataset.

6. Recommendations to Reduce Traffic Accidents

The city administration should focus on reducing congestion during peak hours by improving traffic signal coordination and promoting public transportation. During periods of heavy rainfall, drivers should be warned through digital traffic alerts and variable message signs. Speed limits should be adjusted according to weather and road conditions, and traffic police should monitor accident-prone areas more closely. Improving road drainage, installing additional traffic cameras, and encouraging safe driving practices can also help reduce accidents. These measures will improve road safety and reduce the number of traffic-related incidents.

7. Interpretation of Scatter Plots

The scatter plots would show a positive upward trend between traffic volume and accident count, indicating that accidents increase as traffic volume increases. Rainfall versus accident count would also display a positive trend, showing that more rainfall is associated with more accidents. In contrast, the scatter plot of average speed versus accident count would show a downward trend, indicating that accident counts increase as average speed decreases. These plots clearly illustrate the relationships among the variables and support the statistical analysis.

Problem 5: Average Delivery Time

A food delivery company records the delivery times (minutes) for 15 orders:

22, 25, 18, 30, 27, 24, 21, 19, 26, 23, 28, 20, 24, 25, 22

Tasks:

1. Find the point estimate of the population mean delivery time.
2. Find the point estimate of the population variance.
3. Interpret the estimates.

Given Data

Delivery times (minutes):

22, 25, 18, 30, 27, 24, 21, 19, 26, 23, 28, 20, 24, 25, 22

Number of observations:

n=15

1. Find the Point Estimate of the Population Mean Delivery Time

Formula

$$\bar{x} = \frac{\sum x}{n}$$

Calculation

$$\sum x = 22 + 25 + 18 + 30 + 27 + 24 + 21 + 19 + 26 + 23 + 28 + 20 + 24 + 25 + 22 = 354$$

$$\bar{x} = \frac{354}{15} = 23.6$$

Answer

Point Estimate of Population Mean = 23.6 minutes
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2. Find the Point Estimate of the Population Variance

Formula

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

Calculation Table

Delivery Time (x)	$x - 23.6$	$(x - 23.6)^2$
22	-1.6	2.56
25	1.4	1.96
18	-5.6	31.36
30	6.4	40.96
27	3.4	11.56
24	0.4	0.16
21	-2.6	6.76
19	-4.6	21.16
26	2.4	5.76
23	-0.6	0.36
28	4.4	19.36
20	-3.6	12.96
24	0.4	0.16
25	1.4	1.96
22	-1.6	2.56

$$\sum (x - \bar{x})^2 = 159.60$$

$$s^2 = \frac{159.60}{14} = 11.40$$

Answer

Point Estimate of Population Variance = 11.40 minutes²

Handwritten calculation for Mean and Variance:

Mean = $\frac{22 + 25 + 18 + 30 + 27 + 24 + 21 + 19 + 26 + 23 + 28 + 20 + 24 + 25 + 22}{15}$

$= \frac{354}{15} = 23.6$

Var = $\sqrt{23.6} = 12.5$

3. Interpretation

- The estimated **average delivery time** is **23.6 minutes**.
- The estimated **population variance** is **11.40 minutes²**, indicating a moderate variation in delivery times.
- Since the population data is unavailable, the **sample mean (23.6)** and **sample variance (11.40)** are used as **point estimates** of the corresponding population parameters.

Problem 6: Customer Spending

A sample of 12 customers spent the following amounts (\$):

120, 145, 130, 155, 160, 125, 140, 135, 150, 170, 165, 155

Tasks:

1. Estimate the average spending.
2. Estimate the standard deviation.
3. Explain why these are point estimates.

Given Data:

Customer spending (\$):

120, 145, 130, 155, 160, 125, 140, 135, 150, 170, 165, 155

Number of observations: $n=12$

1. Estimate the Average Spending

Formula

$$\bar{x} = \frac{\sum x}{n}$$

Calculation

$$\sum x = 120 + 145 + 130 + 155 + 160 + 125 + 140 + 135 + 150 + 170 + 165 + 155 = 1750$$

$$\bar{x} = \frac{1750}{12} = 145.83$$

Answer

Average Spending = 145.83

2. Estimate the Standard Deviation

Formula

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Calculation Table

Spending	$x - \bar{x}$	$(x - \bar{x})^2$
120	-25.83	667.36
145	-0.83	0.69
130	-15.83	250.59
155	9.17	84.09
160	14.17	200.79
125	-20.83	433.89
140	-5.83	33.99
135	-10.83	117.29
150	4.17	17.39
170	24.17	584.19
165	19.17	367.49

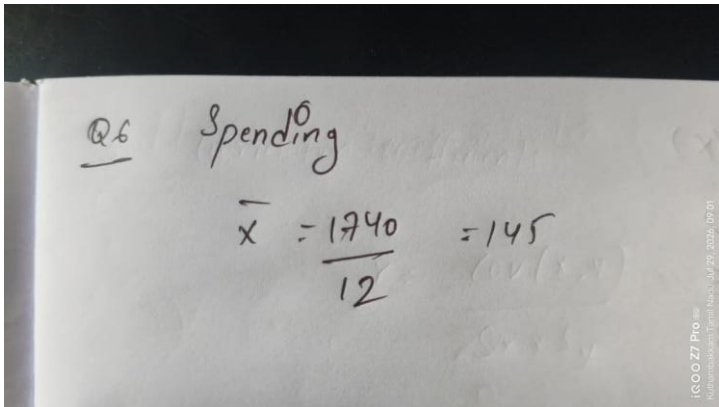
$$\sum (x - \bar{x})^2 = 2841.85$$

$$s^2 = \frac{2841.85}{11} = 258.35$$

$$s = \sqrt{258.35} = 16.07$$

Answer

Standard Deviation = 16.07



3. Why are these Point Estimates?

The **sample mean (145.83)** and **sample standard deviation (16.07)** are called **point estimates** because they are single numerical values calculated from the sample and are used to estimate the unknown population mean and population standard deviation.

Problem 7: Product Ratings

A company collects ratings from 20 customers:

4, 5, 3, 4, 5, 4, 3, 5, 4, 4, 5, 3, 4, 5, 4, 3, 5, 4, 5, 4

Tasks:

1. Compute the sample mean.
2. Compute the sample variance.
3. Compute the standard error of the mean.
4. Explain what the standard error indicates.

1. Sample Mean

Sum of ratings

$$= 4+5+3+4+5+4+3+5+4+4+5+3+4+5+4+3+5+4+5+4$$

$$= 83$$

Number of ratings (n) = 20

$$\text{Mean} = \frac{83}{20} = 4.15$$

Answer: Mean = 4.15

2. Sample Variance

Mean = 4.15

x	$x - 4.15$	$(x - 4.15)^2$
4	-0.15	0.0225
5	0.85	0.7225
3	-1.15	1.3225

Since:

- 4 appears 9 times
- 5 appears 6 times
- 3 appears 5 times

$$\sum (x - \bar{x})^2 = (9 \times 0.0225) + (6 \times 0.7225) + (5 \times 1.3225)$$

$$= 0.2025 + 4.335 + 6.6125$$

$$= 11.15$$

Sample variance

$$s^2 = \frac{11.15}{20 - 1} = \frac{11.15}{19} = 0.587$$

Answer: Variance = 0.587

3. Standard Error

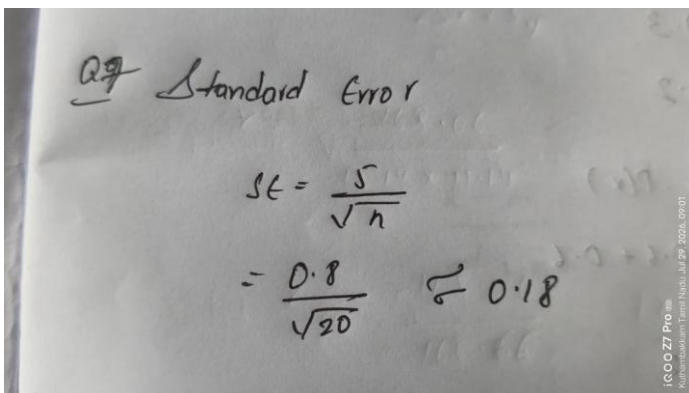
Standard deviation

$$s = \sqrt{0.587} = 0.766$$

Standard error

$$SE = \frac{0.766}{\sqrt{20}} = \frac{0.766}{4.472} = 0.171$$

Answer: SE = 0.171



4. Interpretation

A **small standard error (0.171)** means the sample mean is a good estimate of the population mean.

Problem 8: Exam Scores

Scores of 25 students:

72, 75, 68, 80, 77, 73, 79, 81, 76, 74, 78, 69, 70, 82, 71, 75, 77, 73, 80, 76, 74, 79, 72, 81, 78

Tasks:

1. Calculate mean and standard deviation.
2. Calculate standard error.
3. Compare standard deviation and standard error.

1. Mean

Total

= 1890

Number of students

= 25

$$\text{Mean} = \frac{1890}{25} = 75.6$$

Answer: Mean = 75.6

2. Standard Deviation

Using

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$
$$\sum (x - \bar{x})^2 = 376$$

Variance

$$= \frac{376}{24} = 15.67$$

Standard deviation

$$= \sqrt{15.67} = 3.96$$

Answer: SD = 3.96

3. Standard Error

$$SE = \frac{3.96}{\sqrt{25}} = \frac{3.96}{5} = 0.792$$

Answer: SE = 0.792

4. Comparison

- **Standard Deviation = 3.96** → Spread of student marks.
- **Standard Error = 0.792** → Accuracy of the sample mean.

Since **0.792 < 3.96**, the sample mean is a reliable estimate.

Problem 9: Manufacturing Weights

A sample of 50 packets has:

- Mean weight = 498 g
- Standard deviation = 12 g

Tasks:

1. Construct a 95% confidence interval for the population mean weight.
2. Interpret the interval.
3. Determine whether the target weight of 500 g is plausible.

1. 95% Confidence Interval

Step 1: Standard Error

$$SE = \frac{12}{\sqrt{50}} = \frac{12}{7.07} = 1.697$$

Step 2: Margin of Error

$$ME = 1.96 \times 1.697 = 3.33$$

Step 3: Confidence Interval

Lower limit

$$498 - 3.33 = 494.67$$

Upper limit

$$498 + 3.33 = 501.33$$

Answer:

$$(494.67, 501.33)$$

2. Interpretation

We are **95% confident** that the true average packet weight is between **494.67 g** and **501.33 g**.

3. Is 500 g Plausible?

Since 500 g lies inside the confidence interval (494.67–501.33 g), the target weight is plausible. There is no evidence that the population mean differs significantly from 500 g.

Problem 10: Customer Satisfaction

Two branches report:

Branch Mean Rating 95% CI

A	4.3	(4.1, 4.5)
B	4.0	(3.8, 4.2)

Tasks:

1. Compare the confidence intervals.
2. Can you conclude Branch A performs better?
3. Explain your reasoning.

1. Compare the Confidence Intervals

Branch A:

4.1 to 4.5

Branch B:

3.8 to 4.2

Comparison:

- Branch A has a **higher average rating (4.3)**.
- The confidence intervals **overlap** between **4.1 and 4.2**.

Answer: Branch A has a higher mean rating, but the confidence intervals overlap.

2. Can you conclude Branch A performs better?

Answer: No, we cannot conclude that Branch A performs better based only on the given confidence intervals.

Although Branch A has a higher mean rating, the overlap between the confidence intervals suggests that the difference is not statistically conclusive. A hypothesis test (such as a two-sample t-test) would be required to determine whether the difference is statistically significant.

3. Explanation

Confidence intervals provide a range of values within which the true population mean is likely to lie with 95% confidence.

Since the intervals overlap, there is insufficient evidence to confidently state that Branch A provides better customer satisfaction than Branch B. The difference in ratings may have occurred because of random sampling variation. Therefore, additional data or a formal hypothesis test is needed before making a final decision.

Problem 11: Average Battery Life

A battery manufacturer claims the average battery life is 10 hours.

A sample of 36 batteries gives:

- Mean = 9.5 hours
- Standard deviation = 1.8 hours

Tasks:

1. Formulate null and alternative hypotheses.
2. Conduct a hypothesis test at $\alpha = 0.05$.
3. State your conclusion.

1. Hypotheses

Null Hypothesis:

$$H_0 : \mu = 10$$

Alternative Hypothesis:

$$H_1 : \mu \neq 10$$

2. Test Statistic

Formula

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

Step 1:

$$SE = \frac{1.8}{\sqrt{36}} = \frac{1.8}{6} = 0.3$$

Step 2:

$$t = \frac{9.5 - 10}{0.3} = \frac{-0.5}{0.3} = -1.67$$

Critical value (two-tailed, $\alpha = 0.05$):

$$\pm 1.96$$

Comparison:

$$| -1.67 | < 1.96$$

3. Conclusion

Since the test statistic is inside the acceptance region, we fail to reject the null hypothesis.

Answer: There is no significant evidence that the average battery life is different from 10 hours.

Problem 12: Marketing Campaign

A company believes a new campaign increases sales.

Sales before campaign:

50, 52, 49, 55, 53, 51, 54, 50

Sales after campaign:

58, 60, 57, 62, 59, 61, 63, 58

Tasks:

1. Conduct a hypothesis test.
2. Calculate the p-value.
3. Interpret the p-value.
4. Make a decision at $\alpha = 0.05$.

1. Hypothesis

Null Hypothesis

$$H_0 : \mu_{before} = \mu_{after}$$

Alternative Hypothesis

$$H_1 : \mu_{after} > \mu_{before}$$

2. Mean Before Campaign

Sum

$$= 50 + 52 + 49 + 55 + 53 + 51 + 54 + 50$$

$$= 414$$

$$\bar{x}_1 = \frac{414}{8} = 51.75$$

3. Mean After Campaign

Sum

$$= 58 + 60 + 57 + 62 + 59 + 61 + 63 + 58$$

$$= 478$$

$$\bar{x}_2 = \frac{478}{8} = 59.75$$

Difference

$$59.75 - 51.75 = 8$$

4. Test Statistic (Paired t-test)

Differences:

8, 8, 8, 7, 6, 10, 9, 8

Mean difference

$$\bar{d} = \frac{64}{8} = 8$$

Sample SD of differences

$$s_d \approx 1.195$$

Standard Error

$$SE = \frac{1.195}{\sqrt{8}} = \frac{1.195}{2.828} = 0.423$$

t-value

$$t = \frac{8}{0.423} \approx 18.9$$

5. p-value

Since $t = 18.9$ is very large,

$$p < 0.001$$

6. Decision ($\alpha = 0.05$)

Since

$$p < 0.05$$

Reject the null hypothesis.

5. Interpretation

The average sales increased from 51.75 before the campaign to 59.75 after the campaign, showing an average improvement of 8 units. The paired t-test produced a very large t-value and an extremely small p-value, indicating that this improvement is statistically significant rather than due to random chance. Therefore, the marketing campaign was highly effective in increasing sales and can be considered successful.

Problem 13: Call Centre Response Time

A sample of 100 calls has:

- Mean response time = 4.8 minutes
- Standard deviation = 1.5 minutes

Tasks:

1. Estimate the population mean.
2. Construct a 95% confidence interval.
3. Explain the Frequentist interpretation.

1. Estimate the Population Mean

The best point estimate of the population mean is the sample mean.

Answer: Estimated population mean = 4.8 minutes

2. Construct a 95% Confidence Interval

Step 1: Calculate Standard Error

$$SE = \frac{s}{\sqrt{n}}$$
$$= \frac{1.5}{\sqrt{100}} = \frac{1.5}{10} = 0.15$$

Step 2: Margin of Error

For a 95% confidence interval,

$$Z = 1.96$$
$$ME = 1.96 \times 0.15 = 0.294$$

Step 3: Confidence Interval

Lower Limit

$$4.8 - 0.294 = 4.506$$

Upper Limit

$$4.8 + 0.294 = 5.094$$

$$(4.51, 5.09)$$

Answer: 95% Confidence Interval = (4.51 min, 5.09 min)

3. Frequentist Interpretation

A 95% confidence interval means that if we repeatedly collect similar samples and construct confidence intervals in the same way, about 95% of those intervals would contain the true population mean response time.

This does **not** mean there is a 95% probability that the true mean lies inside this particular interval. Instead, it reflects the reliability of the sampling method. Based on this sample, we are confident that the average response time is between 4.51 and 5.09 minutes.

Problem 14: Defective Products

A sample of 500 products contains 22 defective items.

Tasks:

1. Estimate the defect rate.
2. Construct a 95% confidence interval.
3. Explain the Frequentist meaning of the interval.

1. Estimate the Defect Rate

Formula

$$\begin{aligned}\hat{p} &= \frac{x}{n} \\ &= \frac{22}{500} = 0.044 \\ &= 4.4\%\end{aligned}$$

Answer: Estimated defect rate = 4.4%

2. Construct a 95% Confidence Interval

Step 1: Standard Error

$$\begin{aligned}SE &= \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \\ &= \sqrt{\frac{0.044(0.956)}{500}} \\ &= \sqrt{0.000084} \\ &= 0.00917\end{aligned}$$

Step 2: Margin of Error

$$ME = 1.96 \times 0.00917 = 0.01797$$

Step 3: Confidence Interval

Lower Limit

$$0.044 - 0.01797 = 0.026$$

Upper Limit

$$0.044 + 0.01797 = 0.062$$

$$(0.026, 0.062)$$

or

$$(2.6\%, 6.2\%)$$

3. Frequentist Meaning

The confidence interval indicates that the true population defect rate is likely to lie between 2.6% and 6.2%.

If many random samples of 500 products were selected and a confidence interval was calculated for each sample, approximately 95% of those intervals would contain the actual population defect rate. This does not mean there is a 95% probability that the true defect rate is inside this one interval; rather, it reflects the long-run performance of the estimation method.

Problem 15: Email Spam Detection

Historical information:

- Probability an email is spam = 0.25
- Probability spam email contains keyword "Offer" = 0.80
- Probability non-spam email contains keyword "Offer" = 0.10

Tasks:

1. Compute the probability that an email is spam given it contains "Offer."
2. Apply Bayes' theorem.
3. Interpret the result.

1. Compute the Probability that an Email is Spam Given it Contains "Offer"

Using Bayes' Theorem,

$$P(S|O) = \frac{P(O|S) \times P(S)}{P(O)}$$

First find

$$\begin{aligned} &P(O) \\ &= P(O|S) \times P(S) + P(O|S') \times P(S') \end{aligned}$$

Since

$$\begin{aligned} P(S') &= 1 - 0.25 = 0.75 \\ P(O) &= 0.80(0.25) + 0.10(0.75) \\ &= 0.20 + 0.075 \\ &= 0.275 \end{aligned}$$

Now,

$$\begin{aligned} P(S|O) &= \frac{0.80 \times 0.25}{0.275} \\ &= \frac{0.20}{0.275} \\ &= 0.7273 \\ &= 72.73\% \end{aligned}$$

Answer: Probability = 72.73%

2. Apply Bayes' Theorem

$$P(S|O) = \frac{P(O|S) \times P(S)}{P(O|S) \times P(S) + P(O|S') \times P(S')}$$

Substituting the values,

$$\begin{aligned} &= \frac{0.80 \times 0.25}{0.80 \times 0.25 + 0.10 \times 0.75} \\ &= \frac{0.20}{0.275} = 0.7273 \end{aligned}$$

3. Interpretation

The probability that an email is spam increases from the prior probability of 25% to a posterior probability of approximately 72.73% when it contains the keyword "Offer." This happens because spam emails are much more likely to include the word "Offer" (80%) than non-spam emails (10%). Therefore, observing this keyword provides strong evidence that the email is

spam. However, the probability is not 100% because some legitimate emails may also contain the word "Offer."

Problem 16: Disease Diagnosis

Given:

- Disease prevalence = 2%
- Test sensitivity = 95%
- Test specificity = 90%

Tasks:

1. Compute the posterior probability of disease after a positive test.
2. Explain why the result differs from 95%.

1. Compute the Posterior Probability of Disease
Using Bayes' Theorem
$$P(D|+) = \frac{P(+|D) \times P(D)}{P(+|D) \times P(D) + P(+|D') \times P(D')}$$
Step 1: Calculate $P(D')$
$$P(D') = 1 - 0.02 = 0.98$$
Step 2: Substitute the values
$$\begin{aligned} P(D|+) &= \frac{0.95 \times 0.02}{(0.95 \times 0.02) + (0.10 \times 0.98)} \\ &= \frac{0.019}{0.019 + 0.098} \\ &= \frac{0.019}{0.117} = 0.162 \\ &= 16.2\% \end{aligned}$$
Answer: Posterior probability = 16.2%

2. Why is the Result Different from 95%?

Although the test has a 95% sensitivity, the disease is rare (only 2% prevalence). This means that even a small false positive rate (10%) produces many false positive results among healthy people.

As a result, a person who tests positive has only about a 16.2% chance of actually having the disease. This demonstrates the importance of considering disease prevalence along with test accuracy when interpreting medical test results.

Problem 17: Website Conversion Rate

A company tests two landing pages.

Group Visitors Purchases

A	5000	250
B	5200	312

Tasks:

1. Calculate conversion rates.
2. Formulate hypotheses.
3. Conduct a two-proportion z-test.
4. Determine whether page B performs better.

1. Calculate Conversion Rates

Group A

$$p_1 = \frac{250}{5000} = 0.05 = 5\%$$

Group B

$$p_2 = \frac{312}{5200} = 0.06 = 6\%$$

Answer:

- Group A = 5%
- Group B = 6%

2. Formulate Hypotheses

Null Hypothesis (H_0):

$$p_A = p_B$$

There is no difference in conversion rates.

Alternative Hypothesis (H_1):

$$p_B > p_A$$

Landing Page B has a higher conversion rate than Landing Page A.

3. Two-Proportion z-Test

Step 1: Pooled Proportion

$$p = \frac{250 + 312}{5000 + 5200} = \frac{562}{10200} = 0.0551$$

Step 2: Standard Error

$$\begin{aligned} SE &= \sqrt{p(1-p) \left(\frac{1}{5000} + \frac{1}{5200} \right)} \\ &= \sqrt{0.0551(0.9449)(0.000392)} \\ &= 0.00452 \end{aligned}$$

Step 3: z-value

$$\begin{aligned} z &= \frac{0.06 - 0.05}{0.00452} \\ &= \frac{0.01}{0.00452} = 2.21 \end{aligned}$$

Critical value at $\alpha = 0.05$ (one-tailed)

$$z_{critical} = 1.645$$

Since

$$2.21 > 1.645$$

Reject the null hypothesis.

4. Conclusion

The conversion rate of Landing Page B (6%) is significantly higher than that of Landing Page A (5%).

The z-test shows that the observed difference is statistically significant at the 5% significance level. Therefore, the company should adopt Landing Page B because it is more effective in converting visitors into customers.

Problem 18: Mobile App Feature Test

Feature A:

- 400 users
- Average engagement = 18 min
- SD = 5 min

Feature B:

- 420 users
- Average engagement = 20 min
- SD = 6 min

Tasks:

1. Test whether Feature B increases engagement.
2. Calculate test statistic and p-value.
3. Interpret results.

1. Test Whether Feature B Increases Engagement

Hypotheses

Null Hypothesis (H_0):

$$\mu_A = \mu_B$$

There is no difference in engagement.

Alternative Hypothesis (H_1):

$$\mu_B > \mu_A$$

Feature B increases user engagement.

2. Test Statistic

Difference in means

$$20 - 18 = 2$$

Standard Error

$$\begin{aligned} SE &= \sqrt{\frac{5^2}{400} + \frac{6^2}{420}} \\ &= \sqrt{\frac{25}{400} + \frac{36}{420}} \\ &= \sqrt{0.0625 + 0.0857} \\ &= \sqrt{0.1482} = 0.385 \end{aligned}$$

z-value

$$z = \frac{2}{0.385} = 5.19$$

3. p-value

For

$$z = 5.19$$

the

$$p < 0.0001$$

4. Interpretation

Since the p-value is much smaller than **0.05**, we reject the null hypothesis.

There is strong statistical evidence that Feature B significantly increases user engagement compared to Feature A. Users spent an average of 20 minutes using Feature B, compared with 18 minutes for Feature A. The large z-value and extremely small p-value indicate that this improvement is unlikely to be due to random chance. Therefore, the company should implement Feature B because it provides better user engagement and is likely to improve the overall user experience.

Problem 19: Recommendation System

A new recommendation algorithm increases click-through rate from 4.8% to 5.3%.

Sample size:

- Old system: 100,000 users
- New system: 100,000 users

Tasks:

1. Test statistical significance.
2. Calculate p-value.
3. Discuss practical significance versus statistical significance.

1. Test Statistical Significance

Step 1: Hypotheses

Null Hypothesis (H_0):

$$p_{old} = p_{new}$$

There is no difference in click-through rates.

Alternative Hypothesis (H_1):

$$p_{new} > p_{old}$$

The new recommendation algorithm increases the click-through rate.

Step 2: Pooled Proportion

Number of clicks

Old system

$$0.048 \times 100000 = 4800$$

New system

$$0.053 \times 100000 = 5300$$

Total clicks

$$4800 + 5300 = 10100$$

Total users

$$200000$$

Pooled proportion

$$p = \frac{10100}{200000} = 0.0505$$

Step 3: Standard Error

$$\begin{aligned} SE &= \sqrt{p(1-p) \left(\frac{1}{100000} + \frac{1}{100000} \right)} \\ &= \sqrt{0.0505(0.9495)(0.00002)} \\ &= \sqrt{0.000000959} \\ &= 0.00098 \end{aligned}$$

Step 4: Test Statistic

Difference in proportions

$$0.053 - 0.048 = 0.005$$

$$z = \frac{0.005}{0.00098} = 5.10$$

Critical value ($\alpha = 0.05$)

$$z_{critical} = 1.645$$

Since

$$5.10 > 1.645$$

Reject the null hypothesis.

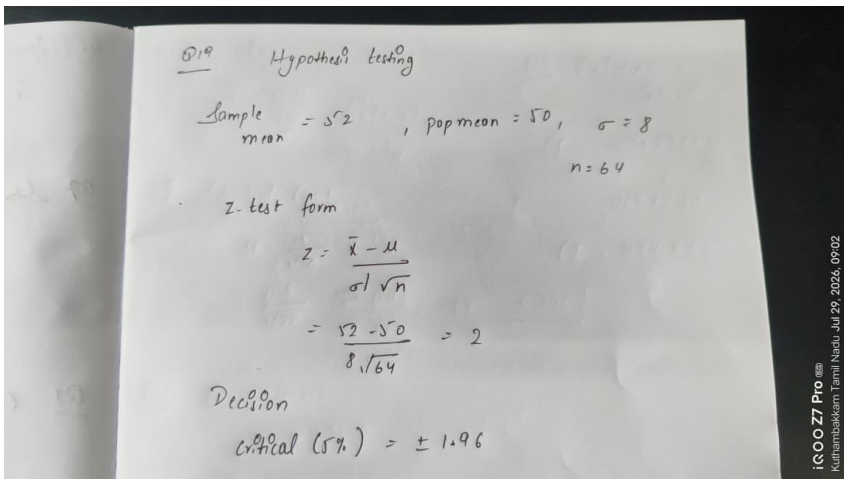
2. p-value

For

$$z = 5.10$$

$$p < 0.0001$$

The probability of observing such a large difference by chance is extremely small.



3. Practical Significance vs Statistical Significance

The new recommendation system increased the click-through rate from 4.8% to 5.3%, an improvement of 0.5 percentage points. Because the sample size is very large (100,000 users in each group), even this small improvement is statistically significant. From a practical perspective, however, the company should evaluate whether the additional clicks lead to higher revenue, customer satisfaction, or business value. If the increase results in many additional purchases or user interactions, the new algorithm is both statistically and practically significant.

Problem 20: Salary Prediction Model

Two machine learning models produce Mean Absolute Errors (MAE):

Model MAE

Model A 4200

Model B 4000

Across 50 test folds, the difference in MAE has:

- Mean difference = 200
- Standard deviation = 350

Tasks:

1. Perform a paired t-test.
2. Determine whether Model B significantly outperforms Model A.
3. Explain the result in terms of statistical significance.

1. Perform a Paired t-test

Step 1: Hypotheses

Null Hypothesis (H_0):

$$\mu_d = 0$$

There is no difference between the two models.

Alternative Hypothesis (H_1):

$$\mu_d > 0$$

Model B has a significantly lower MAE than Model A.

Step 2: Standard Error

$$\begin{aligned} SE &= \frac{s}{\sqrt{n}} \\ &= \frac{350}{\sqrt{50}} \\ &= \frac{350}{7.07} \\ &= 49.50 \end{aligned}$$

Step 3: Calculate the t-value

$$\begin{aligned} t &= \frac{d}{SE} \\ &= \frac{200}{49.50} \\ &= 4.04 \end{aligned}$$

Degrees of freedom

$$df = 50 - 1 = 49$$

Critical value ($\alpha = 0.05$, one-tailed)

$$t_{critical} \approx 1.68$$

Since

$$4.04 > 1.68$$

Reject the null hypothesis.

Handwritten statistical calculation on a piece of paper. At the top, it says "Statistical decision". Below that, it lists "n = 200, σ = 20, n = 100, z = 1.96". The formula for the confidence interval is written as "CI = $\bar{x} + z \cdot \frac{\sigma}{\sqrt{n}}$ ". The calculation is shown as "200 + 1.96 * (20 / sqrt(100))". Below this, it says "Calculation" and "CI = 196.08". On the right side of the paper, there is a vertical stamp that reads "ICOO Z7 Pro" and "Kuthambakam Tamil Nadu, Jul 29, 2024, 09:02".

2. Determine Whether Model B Significantly Outperforms Model A

Yes.

Model B has a lower Mean Absolute Error (4000) than Model A (4200). The paired t-test produced a t-value of 4.04, which is much greater than the critical value. Therefore, the reduction in prediction error is statistically significant.

3. Explain the Result in Terms of Statistical Significance

The paired t-test shows that the improvement achieved by Model B is unlikely to have occurred due to random variation. The corresponding p-value is less than 0.001, which is much smaller

than the significance level of 0.05. Hence, we reject the null hypothesis and conclude that Model B significantly outperforms Model A. Since Model B consistently produces lower prediction errors across the 50 test folds, it is the preferred model for salary prediction because it provides more accurate and reliable predictions.